

20260330 Pseudo EW(H α)

1. Calculation of EW_{pseudo}(H α)

Here we compute a proxy EW(H α) using the observed H α line flux and the broadband R -band flux.

The pseudo equivalent width is defined as

$$\text{EW}_{\text{pseudo}}(\text{H}\alpha) = F_{\text{H}\alpha} / f_{\lambda}(6562.8 \text{ \AA}) \quad (1)$$

where $F_{\text{H}\alpha}$ is the observed H α flux in $[\text{erg s}^{-1} \text{ cm}^{-2}]$, and $f_{\lambda}(6562.8 \text{ \AA})$ is the continuum flux density at H α in $[\text{erg s}^{-1} \text{ cm}^{-2} \text{ \AA}^{-1}]$.

The observed R -band magnitude is stored as `MAGNITUDE_R_UNCORRECTED` in `*_SPATIAL_BINNING_maps_extended.fits`, so we calculate the observed R -band flux in nanomaggy first

$$R_{\text{nanomaggy}} = 10^{(22.5 - m_{R,\text{obs}})/2.5} \quad (2)$$

R -band flux is then converted to flux density per unit frequency (or Jansky) as

$$f_{\nu} = R_{\text{nanomaggy}} \times 3.631 \times 10^{-29} [\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}] \quad (3)$$

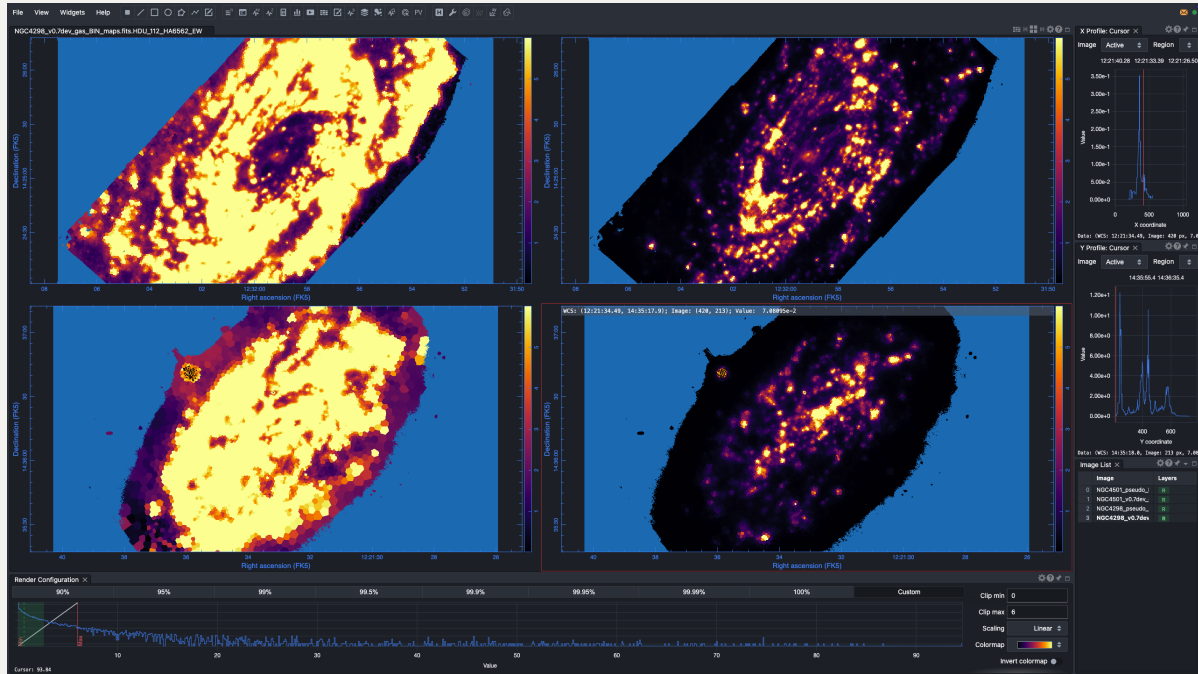
This is then converted to flux density per unit wavelength at 6562.8 $\text{\AA}$ through

$$f_{\lambda} = f_{\nu} c / \lambda^2 [\text{erg s}^{-1} \text{ cm}^{-2} \text{ \AA}^{-1}] \quad (4)$$

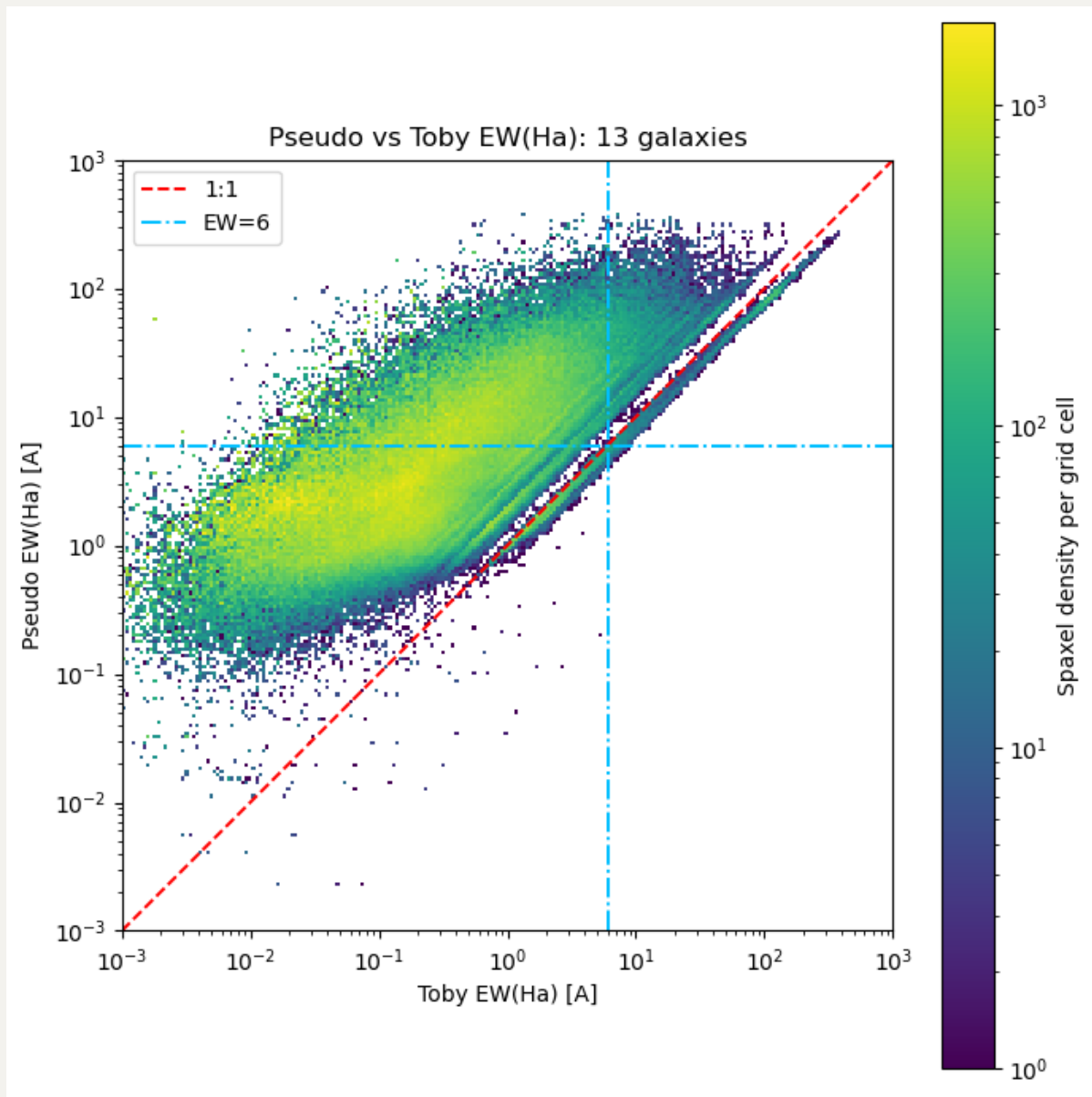
The resulting EW_{proxy}(H α) is therefore expressed in $\text{\AA}$. Again, this quantity is a broadband-based proxy rather than a direct spectroscopic EW measurement.

2. Comparison with Toby's EW(H α)

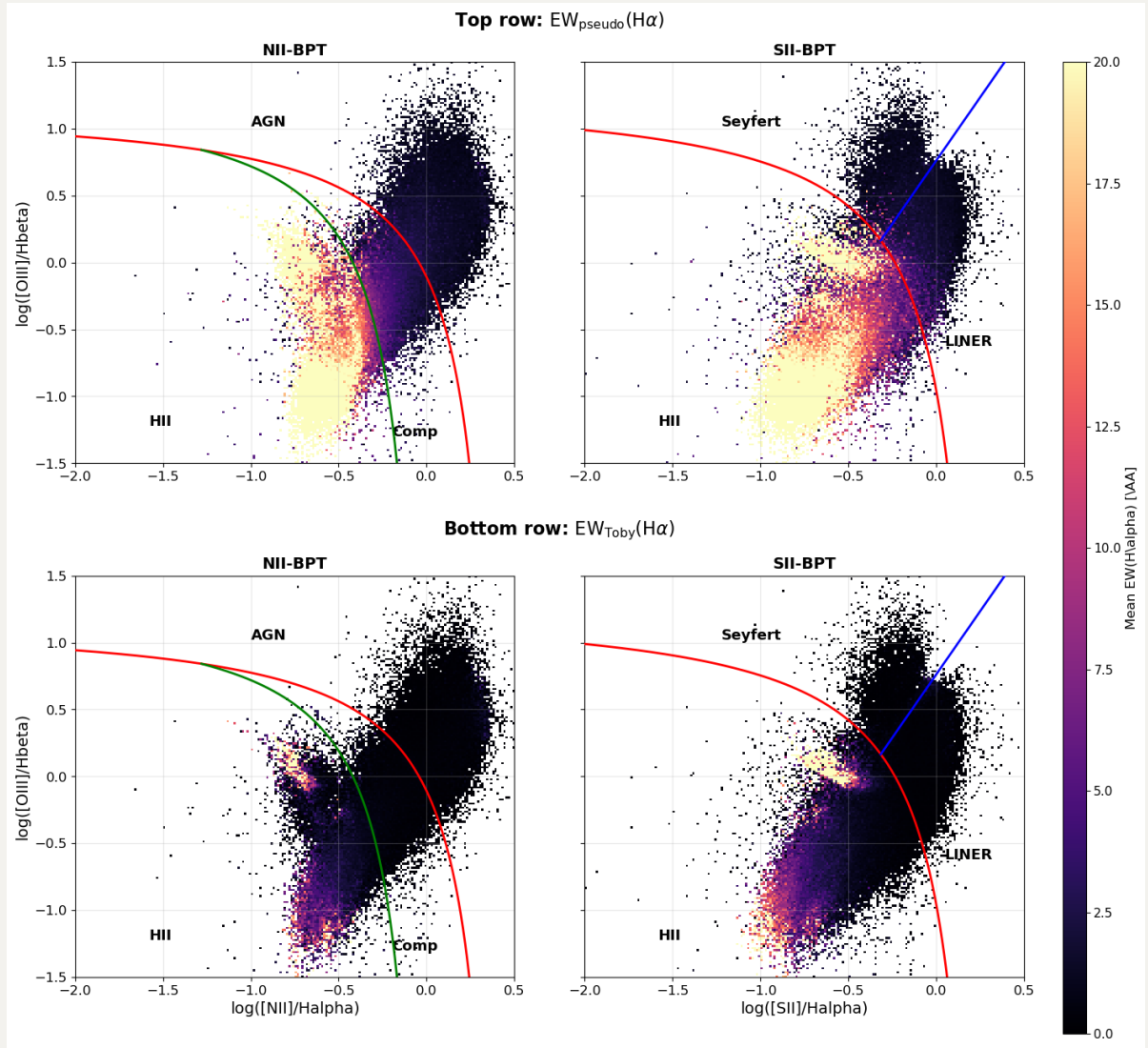
First glance in Carta confirms our idea that Toby's values may be underestimated. Here all the colorbars are set to [0, 6] and the left column is $EW_{\text{pseudo}}(\text{H}\alpha)$ of NGC4501 and NGC4298; while the right column is $EW_{\text{Toby}}(\text{H}\alpha)$. Obviously, Toby's values show much less saturation (SF spaxels).



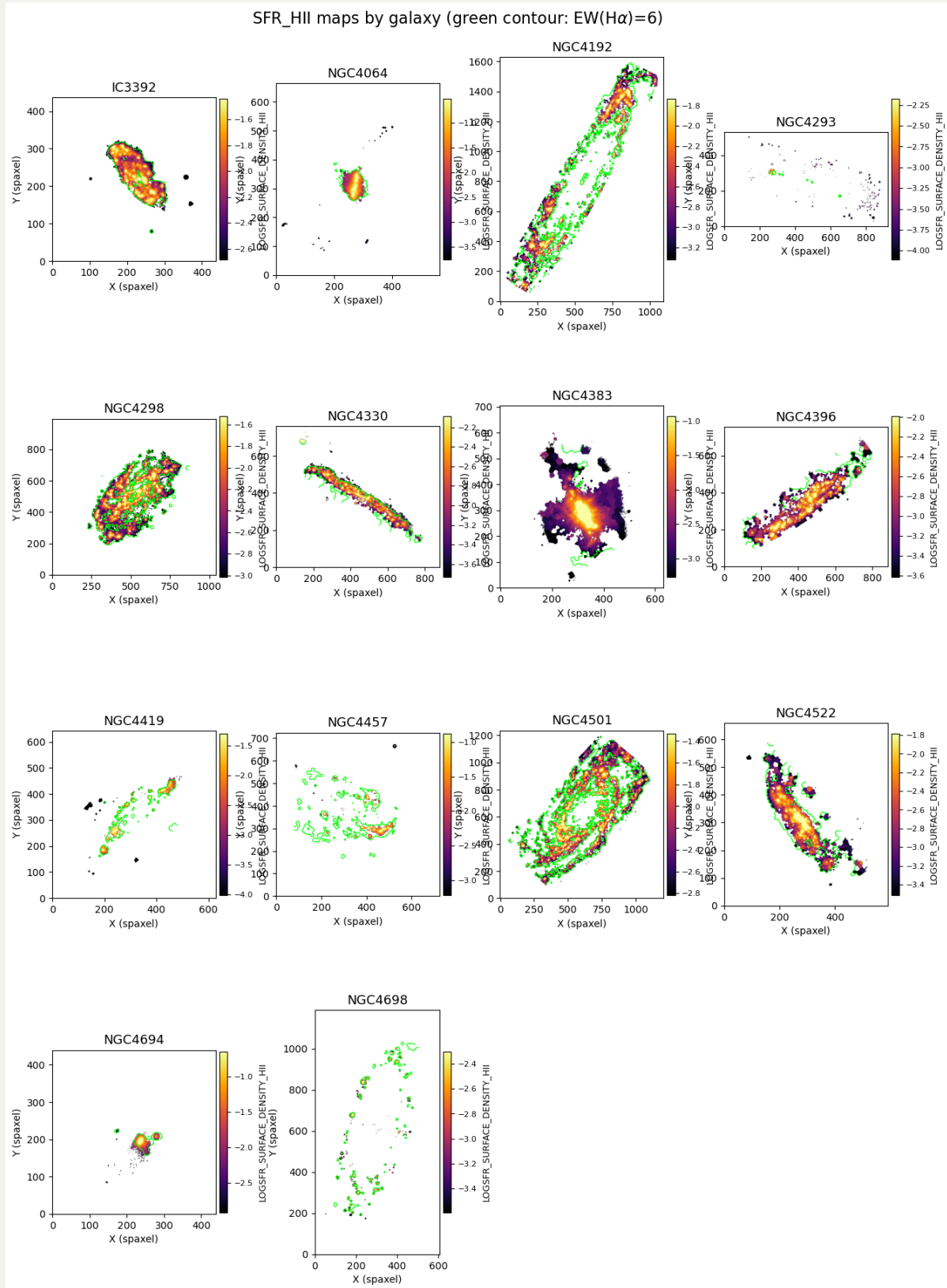
And the one-to-one plot confirms this



Also a comparison on BPT diagrams colocoded by EW(H α)



Clearly, $EW_{\text{pseudo}}(\text{H}\alpha) \geq 6\text{\AA}$ cut mostly trace our previous **HII** selection



3. Plots in paper with new $EW_{\text{pseudo}}(\text{H}\alpha)$

Mostly they remain unchanged visually.

